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Unit – 3 : Session – 9 : SLO – 1 | Queue Implementation Using Linked List

■ I. Enqueue Operation — Insert an Element VAL

- Step 1: Allocate memory for the new node and name it as PTR
- Step 2: SET PTR -> DATA = VAL
- Step 3: IF FRONT = NULL
- SET FRONT = REAR = PTR
- SET FRONT -> NEXT = REAR -> NEXT = NULL
- ELSE
- SET REAR -> NEXT = PTR
- SET REAR = PTR
- SET REAR -> NEXT = NULL
- [END OF IF]
- Step 4: END

■ II. Dequeue Operation — Delete an Element

- Step 1: IF FRONT = NULL
- Write UNDERFLOW
- Go to Step 5
- [END OF IF]
- Step 2: SET PTR = FRONT
- Step 3: SET FRONT = FRONT -> NEXT
- Step 4: FREE PTR
- Step 5: END